OCPP 1.6J Benign Traffic Generation – Findings Log

Started 2026-10-09

1 Goal and scope

- Realistic *benign* OCPP 1.6J traffic (charger ↔ backend) for anomaly detection.
- Benign-only training; anomalies (DoS, billing fraud, idTag misuse, spoofing, injection, protocol violations) only in test data.
- Generator is a means to an end: minimal, every line explainable.

2 Existing datasets – analysis

- **Coventry "OCPP-Centric Hybrid Testbed" (IEEE MEET 2025)** = our `benign_1/2/3.csv`
  - 70 engineered features, no raw payloads.
  - Script-driven: BootNotification every few seconds, ∼4 600 transactions in ∼3 h, only 20–69 Heartbeats.
  - Generator artifact: `connector_id == transaction_id` in every row.
  - Label-leaking columns (`is_anomalous_communication`, `*_zscore`).
- **CICEVSE2024 (UNB)**
  - EVSE-A (real Grizzl-E): TLS to `ocpp.unitedchargers.com`; only sizes/timing visible.
  - Measured: Heartbeat every 60.0 s (60 B out / 90 B in), WebSocket ping every ∼6 s (3.4–10 s), TLS reconnect every 1–5 h.
  - EVSE-B (Raspberry Pi): plaintext ws on port 8765, but permessage-deflate and captures start mid-connection ⇒ not decodable; 1 Hz identical frames, likely not OCPP.
  - Benign: only ∼5 h charging + ∼32 h idle, one charger; no benign pcap for EVSE-B.
  - Attacks only network/host level (nmap scans, hping3 floods, slowloris, cryptojacking, backdoor); no OCPP-level attacks.
- **Federated OCPP 1.6 IDS dataset (Zenodo 14887131, CC-BY)**
  - Only public decodable real-hardware capture: ABB Terra 54 + SteVe, plaintext, no deflate.
  - ARP-spoofing MITM and attack scripts run through the whole capture ⇒ no clean benign period.
  - Measured: Boot interval 60 s; Heartbeat median 59.999 s with second mode 62 s; Heartbeat suppressed while other traffic flows; CP ws ping 30 s, SteVe ping 900 s; reconnect + BootNotification ∼every 4 h; RTT p50 43 ms / p95 108 ms (Wi-Fi).
- **lbbrhzn/ocpp (Home Assistant) GitHub issues**: ∼6 200 real frames, ∼50 vendor/model combos; good for payload structure, weak for timing.
- No public clean, multi-charger, real benign OCPP dataset exists.

3 Literature – data level

- Most OCPP IDS papers: flow features on CICEVSE2024, supervised, only DoS/recon.

- Dalamagkas et al. 2025 (Federated OCPP 1.6): 2 of 4 attacks only detectable with OCPP message features.
- Bean & Manias 2026 survey: encrypted CICEVSE prevents learning application-layer baseline; anomaly detection largely missing.
- Billing fraud, idTag misuse, injection, protocol violations invisible in TLS flows; DoS visible in both.
- Industry (ENCS/ElaadNL, OCA, NIST IR 8473): monitoring at CPO/CSMS via logs and events.
- Jahangir et al. 2024 (IEEE TSG): autoencoder on ACN-Data, session level only.

4 Decisions

- Train on **decoded OCPP messages** (detector at the backend).
- TLS only as ablation (`SSLKEYLOGFILE` capture: encrypted + decrypted view).
- Behaviour from **ACN-Data** (Caltech/JPL/office001): arrivals, dwell, kWh, recurring users.
- Generate offline on a virtual clock (fast); validate a sample live against SteVe.
- Domain randomization: several vendor profiles instead of one "perfect" charger.
- Realism measured with a discriminator test (goal AUC ≈ 0.5).
- Split by ACN day, hold out one site, anomalies only in test (1–5 %).

5 ACN-Data

- Downloaded via API: 66 745 sessions, 115 stations, 2018-04 to 2021-09.
- No 0-kWh sessions ⇒ failed sessions filtered out by ACN; failures must be fabricated.
- `doneChargingTime` partly null; `userID` null for ∼half of Caltech sessions.
- Time-series endpoint: measured charging current + pilot signal every ∼4 s per session.
- Docs say 1 time series per page; tested: 25 complete series per page work.
- Early sessions (Sept 2018) have no time series.
- Download speed limited by ACN server (∼1.7 sessions/s, 25 per page, ∼9 MB per page); run on own server in tmux.
- Download scope: JPL + Caltech Jan–Mar 2019, office001 all of 2019 (∼7 900 sessions).
- Measured curve: implied voltage 203 V (assumed 208 V); no taper, current fluctuates 28–32 A due to ACN adaptive scheduling ⇒ replaced synthetic taper curve.
- Caveat: all curves from a smart-charging site.
- Curve analysis (JPL 4 052 + Caltech 2 549 sessions, run on own server): sample every 4 s; max current bimodal ∼32 A / ∼16 A (car on-board charger), Caltech also 8 A.
- Current starts ∼7 s after plug-in (charging before identification at ACN).
- `doneChargingTime` = last moment current flows (median 0 s); car stays plugged in after full: median 166 min (JPL), 40 min (Caltech).
- Pauses ≥2 min during charging in 13–15 % of sessions ⇒ added SuspendedEVSE (pilot 0) / SuspendedEV → Charging status changes.
- Pilot signal changes median 78× (JPL) / 40× (Caltech) per session; current ≤ pilot in 99 % ⇒ Current.Offered / Power.Offered now from the real pilot.
- Taper at the end (last 10 % < 70 % of median) in 34 % (JPL) / 52 % (Caltech) of sessions.
- Implied voltage per site: 204 V (JPL), 219 V (Caltech) ⇒ bug fixed: readings used fixed 208 V; now per-session voltage.

6 OCPP 1.6J protocol facts used

- Framing: `[2,id,Action,payload]`, `[3,id,payload]`, `[4,id,code,desc,details]`; one outstanding call per direction.
- Schemas from `ocpp` Python library (official JSON schemas).
- SteVe responses (source `CentralSystemService16_Service.java`): Boot → status, currentTime, interval; unknown idTag → `Invalid`; `transactionId` = DB auto-increment.
- Heartbeat interval is set by the backend, not the charger.

7 Vendor profiles (sources)

- ABB Terra AC: boot strings from issue 1275; MeterValueSampleInterval 30 s (ABB manual); measurands Current, Energy, Power, Voltage; `info:"null"`, `vendorErrorCode:"0x0000"`.
- Grizzl-E: model "Grizzl-E Smart Connect" (22 chars, violates schema max 20 on the real device); `meterType` junk values; `location:"Body"`, `phase:"L1-N"`; timestamps end in `.001Z`.
- Wallbox Pulsar Plus: Faulted/InternalError on boot then Available; connector-0 clock-aligned energy `"0.0"` outside transactions.
- Other vendor defaults (Zaptec, Easee, Enel X, Alpitronic): Heartbeat 60–86 400 s, MeterValues 0–90 s.

8 Generator (implementation)

- `fetcher.py`: ACN API download (sessions and time series), resumable.
- `profiles.py`: 3 vendor profiles, every value sourced or marked ASSUMPTION.
- `generator.py`: per charger 4 steps – calls, heartbeats, outages, wire (latency + SteVe responses).
- `validate.py`: every frame against official schemas; order; meter monotonic.
- JPL March 2019: 52 chargers, ∼5 M frames, ∼1 min runtime.
- Bug found and fixed: Heartbeat sneaked in before every MeterValues (60 s vs 60 s); rule: no Heartbeat if a message is due within 1 s.
- Measured ACN current replaces the synthetic taper curve where a time series exists; synthetic curve kept only as fallback.
- Calibration applied (v2): per-vendor session order, session-relative energy (ABB, Grizzl-E), vendor message-ID styles, vendor timestamp formats, Wallbox app starts via RemoteStartTransaction, Grizzl-E free-vend with own serial, Wallbox 8-measurand readings and clock-aligned lifetime readings, server time 40 ms (was 3 ms) and network 1 ms (was 43 ms), clock offset 0–0.35 s (was ±2 s), Grizzl-E heartbeat without 62 s mode, re-sent status keeps old timestamp.
- Artefact found and fixed: ACN times are whole seconds ⇒ plug-in/unplug frames arrived at exactly `.000`; added sub-second offset.
- Validation v2: JPL March 2019, 12 chargers, ∼1.1 M frames, 0 schema or consistency problems (Grizzl-E 05.x boot is schema-valid).
- Code size: generator 401, profiles 119, validate 53, fetcher 121 lines.

9 Calibration from real captures (ABB Terra 54, Grizzl-E)

- Method: tshark per TCP stream; request/response matched by message ID; network vs. server time split via CSMS empty TCP ACK.
- Capture clock trustworthy: SteVe currentTime within −25 ms over 2.3 days.

- **Reboot**: ABB reboots on a strict 4 h 01 m schedule (∼6/day, 21 intervals 4.015–4.053 h) – possibly operator-configured; replaces "1 per 14 days".
- Reboot sequence (ABB): Unavailable on connectors 0–3 → charger closes WebSocket → offline 150 ± 2 s → Boot → status within 2–7 s → transient Faulted/OtherError (+11 s, +20 s) → Available (+31 s).
- Queued StopTransaction sent as 2nd message after reboot, keeps original timestamp (153 s old).
- Reconnect without reboot: status for all connectors re-sent with original (hours-old) timestamps.
- Grizzl-E: 6 server-initiated closes in 36.6 h, bursty; reconnect after 10 s (FIN) or 1 s (RST); Boot ∼3.7 s after SYN; Boot + 2 status on every reconnect.
- **Clock**: ABB sawtooth drift +25 ppm (∼+0.33 s after 4 h), reset at boot; clock taken from BootNotification.conf currentTime.
- **Heartbeat**: ABB 85 % 60.0 s / 15 % 62.0 s confirmed; Grizzl-E 60.0 s ±0.4 s (no 62 s mode).
- Heartbeat timer reset only by the charger's own requests (n=2445).
- **Latency correction**: network RTT only ∼1 ms; the 43 ms median is SteVe processing time (Heartbeat p50 39/p95 100 ms, StartTransaction 105 ms).
- MeterValues (ABB DC): every 10.0 s, only in transactions; Energy (integer Wh) + SoC; Transaction.Begin/End as separate MeterValues.
- Message IDs (ABB): `"<bootCounter>:<seq>"`; seq restarts at Boot ⇒ uuid4 everywhere is a fingerprint.
- WebSocket pings: ABB 30.0 s; Grizzl-E and server each 15 s, offset ∼10 s.
- Not usable: ABB attack periods (gaps/short connections), Grizzl-E MeterValues (no charging traffic in pcap).

10 Calibration from real charger logs (Home Assistant OCPP issues)

- Sources: issues #176, #442, #880, #906, #994, #1069, #1684, #1717, #1742, #1756, #1767 (+ attachments); ABB 11 chargers / 7 sessions, Grizzl-E 6 / 2, Wallbox 14 / 5.
- Caveat: most sessions remote-started by Home Assistant; intervals (heartbeat 3600, sample, clock) are set by HA, not the charger.
- No real card-tap timing found ⇒ 15 s stays an assumption.
- **Session order differs per vendor**: ABB sends Charging 0.8 s *before* StartTransaction and Finishing before StopTransaction; Wallbox Start → Charging +4–7 s, Finishing → Stop → Available +20–50 s; Grizzl-E Start → Charging +6–7 s.
- SuspendedEV when the car is full: not observed in these logs (kept as assumption; supported by ACN `doneChargingTime`).
- **Session-relative energy**: ABB and Grizzl-E send `meterStart: 0` and count up per session; Wallbox reports the lifetime register.
- MeterValues interval ABB 60 s (not 30 s), Grizzl-E 60 s; ABB sends a separate Transaction.Begin MeterValues (energy 0) 3–10 s after Start.
- Wallbox measurands: Current L1/L2/L3 ("0.0" on unused phases), Energy register + Interval, Power, Power.Offered, Frequency (no unit); StopTransaction always `transactionData: []`.
- Wallbox clock-aligned readings on connectors 0 and 1, lifetime value, timestamp = start of interval (sent at its end); "0.0" only right after boot.
- Wallbox Faulted on boot only on some boots (1 of 3 logs).
- **Message IDs**: ABB 7-digit fixed sequence repeating after every boot; Grizzl-E (05.x) serial + 8 hex; Wallbox random decimal up to 2^{31}.
- Timestamps: ABB `.000Z`; Grizzl-E 05.x `.001Z`; Wallbox whole seconds.
- Grizzl-E profile mixed two firmware generations (GWM-07 boot, 05.x message format) ⇒ switched to consistent 05.x generation (model `GRS-*`, schema-valid).

- Grizzl-E free-vend idTag = its own serial, Authorize 1–2 s after plug-in.
- Not modelled: ABB SecurityEventNotification at boot (SteVe lacks security extensions), ABB 4 h reboot cycle (likely operator config of the DC unit).

11 Attack (test) data

- Public attack data: CICEVSE2024 network/host attacks only (TLS, not convertible to OCPP messages); Federated dataset has 4 decodable OCPP attacks; Coventry only engineered features; no public billing-fraud or spoofing traffic.
- Converting foreign attack captures into our format would let the model learn "foreign environment" instead of "attack" ⇒ attacks are injected through the same generator (same profiles, IDs, timing); only the manipulation differs.
- Optional later: Federated attacks as separate cross-environment test, reported as such.
- `attacks.py`: billing fraud (energy/power/current and meterStop scaled to 50–80 %), cloned card (victim's tag at another idle charger during victim's session, curve copied from a real session), DoS flood (Heartbeat/Status at 5–20/s for 1–5 min), spoofing (second device, other vendor fingerprint, same charger ID, 10–60 min), replay (old StopTransaction re-sent 1–24 h later).
- Every frame labelled (`benign` or attack name); attacks are schema-valid OCPP.
- Test week JPL 2019-03-15..21: 14 attacks, ∼1.08 M frames, 1.2 % malicious frames (DoS 10 226, billing fraud 1 214, cloned card 968, spoofing 336, replay 10).
- Scope: closed world – all vendors seen in training; new-vendor generalisation (leave-one-vendor-out) as optional experiment.

12 Features, model, infrastructure

- Data stays on own server (tarulo): ACN time series downloaded there; time-series files also serve as session source (contain all session fields).
- Bug found: some sessions have measured current but no pilot signal ⇒ keep current, offered current 32 A.
- Split by days (JPL 2019): train Jan 1–Feb 28 (benign), val Mar 1–7 (benign, threshold), test Mar 8–31 (benign + ∼2 attacks/day).
- `features.py`: only requests (charger + backend calls); SteVe answers dropped. Per request: token (action + status / meter context / error flag) and 9 numbers: $\log(1 + \Delta t)$, energy delta (kWh), power (kW), current (A/32), offered current (A/32), session energy at stop, idTag active at another charger, boot fingerprint differs from registry, transactionId not open on this charger.
- Vendor format removed (message IDs, timestamp style, info texts, units, phase layout) ⇒ model sees behaviour, not vendor.
- Windows of 64 requests per charger, stride 16; window labelled malicious if any message is.
- Model (`train.py`): LSTM autoencoder (embedding 16, hidden 64), loss = token cross-entropy + MSE of standardised numbers; trained on benign only.
- Threshold = 99.5th percentile of benign validation scores; metrics per attack type: recall, ROC-AUC; overall FPR and precision. Labels used for evaluation only.
- Training on AAU AI-LAB (Slurm, partition `l4`, NVIDIA L4, container `pytorch_26.09.sif`); access via AAU VPN (openconnect) + SSH key.
- Smoke test (1 epoch, 12 chargers, 1 week): pipeline works end-to-end on GPU; numbers not meaningful yet.
- Dataset v1 (JPL 2019, 52 chargers): train 281 737 / val 33 221 / test 115 746 windows; 48 attacks in test, 1 926 attack windows (1.7 %); features only 29 MB compressed.

- **Run 1** (job 910981, 10 epochs, L4, minutes): train loss 0.80 → 0.49; threshold 99.5th pct of val; test FPR 0.57 %, precision 2.1 %.
- Run 1 ROC-AUC: replay 0.99, DoS 0.98, spoofing 0.82, cloned card 0.78, billing fraud 0.70.
- Run 1 recall at threshold ≈ 0 (replay 11 %, cloned card 3 %, others 0 %): scores rank attacks high but the threshold sits above them.
- Diagnosis: window score = mean error over 64 positions × 9 numbers ⇒ a single anomalous message (e.g. one flag) is diluted (∼1/576); rare benign events (reboots, faults, outages) dominate the benign tail.
- **Run 2** (job 910985): score = max error of a single message; never-firing flags = $10\,\sigma$; event-level metrics added. Replay 11/11 events detected (AUC 0.999); FPR 0.50 %, precision 7.8 %; other attacks 0–1 of 8–9 events; threshold 8 295 (abnormally high).
- Diagnosis run 2: (a) sparse feature "session energy at stop" (non-zero in 0.06 %) had tiny std ⇒ normal stops = $122\,\sigma$; (b) "transaction not open" flag fired 2 080× in benign train: Wallbox clock-aligned readings report a past interval and keep the transactionId after the session ended.
- Fixes for run 3: sparse features scaled by their non-zero values (0 stays 0); transaction flag ignores clock-aligned readings.
- **Run 3** (job 910995): threshold 77 (was 8 295); FPR 0.37 %, precision 17.7 %. Events detected: replay 11/11, cloned card 5/8, spoofing 5/8, DoS 2/9, billing fraud 0/8. AUC: replay 1.00, DoS 0.97, spoofing 0.87, cloned card 0.80, billing fraud 0.72.
- Diagnosis run 3: (a) spoofing missed when the charger never booted during training (no registered fingerprint); (b) cloned card missed when the victim charger is a Grizzl-E (uses its own serial, so the victim's card is never "active") – attack design bug; (c) DoS is a rate pattern, a single fast Heartbeat is only $\sim 7\,\sigma$; (d) billing fraud scales energy, power and current consistently ⇒ looks plausible per message (expected; needs a second data source).

13 Cross-environment evaluation (Federated OCPP 1.6 dataset)

- `federated.py`: real ABB Terra 54 + SteVe pcaps → our JSONL; MITM duplicates removed; several messages per TCP packet split. Attack 1 (charging-profile manipulation): 19 279 frames, 6 717 SetChargingProfile = attack. Attack 2 (denial of charge): 65 956 frames, attacker RemoteStarts + the sessions they start = attack.
- Run-3 model evaluated without retraining (`train.py --model`).
- Window level not usable: attacker active every ∼30 s ⇒ every window contains an attack, no benign windows (FPR undefined).
- `crosseval.py`: per-message scores (max over windows containing the message), same threshold.
- Attack 1: benign requests flagged 5.8 %, attack requests 1.0 %, AUC 0.40.
- Attack 2: benign requests flagged 1.0 %, attack requests 18.0 %, AUC 0.47; most flags on MeterValues/Transaction.End (a message type our profiles never produce).
- Finding: the model does *not* flag the foreign environment wholesale (benign 1–6 % vs. 0.4 % in-domain), but does not detect these attacks either (AUC ≈ random).
- Reasons: (a) attack types not modelled (backend commands SetChargingProfile / RemoteStart are outside our charger-behaviour features); (b) unseen request types (SetChargingProfile, Reset, RemoteStop, Transaction.End) only add token cross-entropy (20), far below the threshold (77) set by numeric errors ⇒ novelty of message types is under-weighted.
- Conclusion: in-domain results do not transfer to other attack types and environments; supports closed-world scope statement and motivates a "request type never seen" feature.

14 Remaining assumptions

- Rare-event rates (outage 1/7 d, reboot 1/14 d, fault 1/30 d, wrong card 3 %) and fault error codes – no benign source.
- Card-tap / app-start time (∼15 s); free-vend tag for ABB/Wallbox (charger serial, weak evidence).
- Charger answer time to RemoteStart (0.1–0.5 s); Available delay ABB and Grizzl-E.
- SuspendedEV when the car is full (not seen in logs; implied by ACN `doneChargingTime`).
- Network latency of a LAN (1 ms); a cellular/WAN deployment would be slower.
- Wallbox heartbeat jitter.

15 Open

- Real benign data: own wallbox + SteVe; ask Federated/Coventry authors; companies contacted; AAU faculties.
- `live.py` against SteVe; discriminator test.